

MATH 350-2 Advanced Calculus

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Outline

- 1 Real Analysis Lecture 14
 - Complete metric spaces
 - Limit of a function
 - Continuity

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Complete spaces

Theorem

Let (M, d) be a metric space.

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Let (M, d) be a metric space. Then for any compact subset $S \subseteq M$, the subspace (S, d) is complete.

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Let (M, d) be a metric space and $S \subseteq M$ compact.

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Let (M, d) be a metric space and $S \subseteq M$ compact.
To show that (S, d) is complete, we must show that every Cauchy sequence in S converges to a value in S .

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Suppose that $\{x_n\}$ is a Cauchy sequence in (S, d) .

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Suppose that $\{x_n\}$ is a Cauchy sequence in (S, d) .

If the set X of all values of $\{x_n\}$ is finite, then the Cauchy condition implies $\{x_n\}$ converges.

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Assume instead X is infinite.

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TIME OUT!



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Lemma

Let (M, d) be a metric space and let $S \subseteq M$ be compact.

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Let (M, d) be a metric space and let $S \subseteq M$ be compact. Then for any infinite subset $X \subseteq S$, the set S has an accumulation point of X .

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To see this, suppose otherwise.

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Proof.

To see this, suppose otherwise.

Then for all $x \in S$, there exists $r_x > 0$ such that $B_S(x, r_x) \cap X$ is empty if $x \notin X$ or equal to $\{x\}$ if $x \in X$.

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Then for all $x \in S$, there exists $r_x > 0$ such that $B_S(x, r_x) \cap X$ is empty if $x \notin X$ or equal to $\{x\}$ if $x \in X$.

The open cover $\{U_i : i \in I\}$ with $I = S$ and $U_i = B_S(i; r_i)$ of S has no finite subcover.

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Let (M, d) be a metric space and let $S \subseteq M$ be compact. Then for any infinite subset $X \subseteq S$, the set S has an accumulation point of X .

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To see this, suppose otherwise.

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The open cover $\{U_i : i \in I\}$ with $I = S$ and $U_i = B_S(i; r_i)$ of S has no finite subcover.

This is a contradiction, so S has an accumulation point of X . □

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Let L be an accumulation point of X in S .

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Let L be an accumulation point of X in S .

Let $\epsilon > 0$.

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Proof.

Let L be an accumulation point of X in S .

Let $\epsilon > 0$.

Then $\exists N > 0$ such that $m, n \geq N$ implies $d(x_m, x_n) < \epsilon/2$.

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Let L be an accumulation point of X in S .

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Moreover, $B_S(L, \epsilon/2)$ contains infinitely many points of X .

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Hence it contains x_m for some $m \geq N$.

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Hence it contains x_m for some $m \geq N$.

It follows that for all $n \geq N$,

$$d(x_n, L) \leq d(x_n, x_m) + d(x_m, L) < \epsilon/2 + \epsilon/2 = \epsilon.$$

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Since $\epsilon > 0$ was arbitrary, this proves $\{x_n\}$ converges to L . □ ↻ 🔍

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Limit of a function

Let (S, d_S) and (T, d_T) be metric spaces and $A \subseteq S$.

Limit of a function

Let (S, d_S) and (T, d_T) be metric spaces and $A \subseteq S$. Also, let $f : A \rightarrow T$ be a function.

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Then for any accumulation point p of A and element $L \in T$, we say that the **limit as x approaches p of $f(x)$ is L**

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$$\lim_{x \rightarrow p} f(x) = L$$

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if for all $\epsilon > 0$ there exists $\delta > 0$ such that

$$0 < d_S(x, p) < \delta \Rightarrow d_T(f(x), L) < \epsilon.$$

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if for all $\epsilon > 0$ there exists $\delta > 0$ such that

$$x \in B_S(p; \delta) \setminus \{p\} \Rightarrow f(x) \in B_T(L; \epsilon).$$

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$$\lim_{x \rightarrow p} f(x) = L$$

if for all $\epsilon > 0$ there exists $\delta > 0$ such that

$$f(B_S(p; \delta) \setminus \{p\}) \subseteq B_T(L; \epsilon).$$

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Limit of a function

Theorem

Let (S, d_S) and (T, d_T) be metric spaces, $A \subseteq S$, and $f : A \rightarrow T$ be a function. Then

$$\lim_{x \rightarrow a} f(x) = L$$

if and only if

$$\lim_{n \rightarrow \infty} f(x_n) = L$$

for all sequences $\{x_n\}$ of values in $A \setminus \{a\}$ which converge to a .

Limit of a function

Proof.

($\Rightarrow$) Suppose that $\lim_{x \rightarrow a} f(x) = L$ and let $\{x_n\}$ be a sequence of values in $A \setminus \{a\}$ which converges to a .

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Then there exists $\delta > 0$ such that $0 < d_S(x, a) < \delta$ implies $d_T(f(x), L) < \epsilon$.

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($\Rightarrow$) Suppose that $\lim_{x \rightarrow a} f(x) = L$ and let $\{x_n\}$ be a sequence of values in $A \setminus \{a\}$ which converges to a .

Let $\epsilon > 0$.

Then there exists $\delta > 0$ such that $0 < d_S(x, a) < \delta$ implies $d_T(f(x), L) < \epsilon$.

Also, there exists $N \in \mathbb{Z}_+$ such that $n \geq N$ implies $d_S(x_n, a) < \delta$ for all $n \geq N$.

Limit of a function

Proof.

($\Rightarrow$) Suppose that $\lim_{x \rightarrow a} f(x) = L$ and let $\{x_n\}$ be a sequence of values in $A \setminus \{a\}$ which converges to a .

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Then there exists $\delta > 0$ such that $0 < d_S(x, a) < \delta$ implies $d_T(f(x), L) < \epsilon$.

Also, there exists $N \in \mathbb{Z}_+$ such that $n \geq N$ implies $d_S(x_n, a) < \delta$ for all $n \geq N$.

Therefore since $x_n \neq a$ for all n , we have that

$$n \geq N \Rightarrow 0 < d_S(x_n, a) < \delta \Rightarrow d_T(f(x_n), L) < \epsilon.$$

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$$n \geq N \Rightarrow 0 < d_S(x_n, a) < \delta \Rightarrow d_T(f(x_n), L) < \epsilon.$$

Hence for all $n \geq N$, we know $d_T(f(x_n), L) < \epsilon$.

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($\Rightarrow$) Suppose that $\lim_{x \rightarrow a} f(x) = L$ and let $\{x_n\}$ be a sequence of values in $A \setminus \{a\}$ which converges to a .

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Hence for all $n \geq N$, we know $d_T(f(x_n), L) < \epsilon$.

Since $\epsilon > 0$ was arbitrary, this proves $\lim_{n \rightarrow \infty} f(x_n) = L$. □

Limit of a function

Proof.

($\Leftarrow$) Suppose that for every sequence $\{x_n\}$ of values in $A \setminus \{a\}$ which converges to a , we have $\lim_{n \rightarrow \infty} f(x_n) = L$.

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($\Leftarrow$) Suppose that for every sequence $\{x_n\}$ of values in $A \setminus \{a\}$ which converges to a , we have $\lim_{n \rightarrow \infty} f(x_n) = L$. We must show that $\lim_{x \rightarrow a} f(x) = L$.

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To see this, we assume otherwise.

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To see this, we assume otherwise.

Then there exists $\epsilon > 0$ such that for all $\delta > 0$ there exists x with $0 < d_S(x, a) < \delta$ and $d_T(f(x), L) \geq \epsilon$.

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($\Leftarrow$) Suppose that for every sequence $\{x_n\}$ of values in $A \setminus \{a\}$ which converges to a , we have $\lim_{n \rightarrow \infty} f(x_n) = L$. We must show that $\lim_{x \rightarrow a} f(x) = L$.

To see this, we assume otherwise.

Then there exists $\epsilon > 0$ such that for all $\delta > 0$ there exists x with $0 < d_S(x, a) < \delta$ and $d_T(f(x), L) \geq \epsilon$.

For each $n \in \mathbb{Z}_+$, take $x_n \in A$ with $0 < d_S(x_n, a) < 1/n$ and $d_T(f(x_n), L) \geq \epsilon$.

Limit of a function

Proof.

($\Leftarrow$) Suppose that for every sequence $\{x_n\}$ of values in $A \setminus \{a\}$ which converges to a , we have $\lim_{n \rightarrow \infty} f(x_n) = L$. We must show that $\lim_{x \rightarrow a} f(x) = L$.

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For each $n \in \mathbb{Z}_+$, take $x_n \in A$ with $0 < d_S(x_n, a) < 1/n$ and $d_T(f(x_n), L) \geq \epsilon$.

In this case, $\{x_n\}$ is a sequence of values in $A \setminus \{a\}$ which converges to a .

Limit of a function

Proof.

($\Leftarrow$) Suppose that for every sequence $\{x_n\}$ of values in $A \setminus \{a\}$ which converges to a , we have $\lim_{n \rightarrow \infty} f(x_n) = L$. We must show that $\lim_{x \rightarrow a} f(x) = L$.

To see this, we assume otherwise.

Then there exists $\epsilon > 0$ such that for all $\delta > 0$ there exists x with $0 < d_S(x, a) < \delta$ and $d_T(f(x), L) \geq \epsilon$.

For each $n \in \mathbb{Z}_+$, take $x_n \in A$ with $0 < d_S(x_n, a) < 1/n$ and $d_T(f(x_n), L) \geq \epsilon$.

In this case, $\{x_n\}$ is a sequence of values in $A \setminus \{a\}$ which converges to a .

However, $\{f(x_n)\}$ never gets closer than ϵ to L , so it cannot converge to L .

Limit of a function

Proof.

($\Leftarrow$) Suppose that for every sequence $\{x_n\}$ of values in $A \setminus \{a\}$ which converges to a , we have $\lim_{n \rightarrow \infty} f(x_n) = L$. We must show that $\lim_{x \rightarrow a} f(x) = L$.

To see this, we assume otherwise.

Then there exists $\epsilon > 0$ such that for all $\delta > 0$ there exists x with $0 < d_S(x, a) < \delta$ and $d_T(f(x), L) \geq \epsilon$.

For each $n \in \mathbb{Z}_+$, take $x_n \in A$ with $0 < d_S(x_n, a) < 1/n$ and $d_T(f(x_n), L) \geq \epsilon$.

In this case, $\{x_n\}$ is a sequence of values in $A \setminus \{a\}$ which converges to a .

However, $\{f(x_n)\}$ never gets closer than ϵ to L , so it cannot converge to L .

This is a contradiction.



Example

Consider the function

$$f : \mathbb{R} \rightarrow \mathbb{R}^2$$

between metric spaces with the Euclidean metric defined by

$$f(t) = (1 - t, t^2).$$

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$$\lim_{t \rightarrow 0} f(t) = (1, 0).$$

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To see this, let $\epsilon > 0$.

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To see this, let $\epsilon > 0$. Choose $\delta > 0$ by $\delta = \min\{1, \epsilon/\sqrt{2}\}$.

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To see this, let $\epsilon > 0$. Choose $\delta > 0$ by $\delta = \min\{1, \epsilon/\sqrt{2}\}$. Then for $d_{\mathbb{R}}(t, 0) < \delta$, we have

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To see this, let $\epsilon > 0$. Choose $\delta > 0$ by $\delta = \min\{1, \epsilon/\sqrt{2}\}$. Then for $d_{\mathbb{R}^2}(f(t), (1, 0)) < \delta$, we have

$$d_{\mathbb{R}^2}(f(t), (1, 0)) = \sqrt{(1 - (1 - t))^2 + (0 - t^2)^2} < |t|\sqrt{1 + t^2} \leq |t|\sqrt{2} \leq \epsilon$$

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To see this, let $\epsilon > 0$. Choose $\delta > 0$ by $\delta = \min\{1, \epsilon/\sqrt{2}\}$. Then for $d_{\mathbb{R}^2}(f(t), (1, 0)) < \delta$, we have

$$d_{\mathbb{R}^2}(f(t), (1, 0)) = \sqrt{(1 - (1 - t))^2 + (0 - t^2)^2} < |t|\sqrt{1 + t^2} \leq |t|\sqrt{2} \leq \epsilon$$

Since $\epsilon > 0$ was arbitrary, we are done.

Challenge!

Consider the function

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

between metric spaces with the Euclidean metric defined by

$$f(x, y) = \frac{x^2 y}{x^4 + y^2}.$$

Challenge!

Consider the function

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Question

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between metric spaces with the Euclidean metric defined by

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Question

What is the limit as $(x, y) \rightarrow (0, 0)$?

Challenge!

Consider the sequence $\{(x_n, y_n)\}$ defined by

$$(x_n, y_n) = (1/n, 0).$$

Challenge!

Consider the sequence $\{(x_n, y_n)\}$ defined by

$$(x_n, y_n) = (1/n, 0).$$

Then

$$\lim_{n \rightarrow \infty} (x_n, y_n) = \lim_{n \rightarrow \infty} (1/n, 0) = (0, 0)$$

Challenge!

Consider the sequence $\{(x_n, y_n)\}$ defined by

$$(x_n, y_n) = (1/n, 0).$$

Then

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and

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The limit DOES NOT EXIST!

Outline

- 1 Real Analysis Lecture 14
 - Complete metric spaces
 - Limit of a function
 - Continuity

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Equivalently, $f(B_S(p; \delta)) \subseteq B_T(f(p); \epsilon)$. We say f is continuous on $A \subseteq S$ if $f(x)$ is continuous at every point $p \in A$. If f is continuous on S , we simply say f is continuous.

Sequence version

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Intuitively, continuity means

$$\lim_{n \rightarrow \infty} f(x_n) = f\left(\lim_{n \rightarrow \infty} x_n\right).$$

Compositions

Theorem

Suppose (S, d_S) , (T, d_T) , and (U, d_U) are metric spaces and $f : S \rightarrow T$ and $g : T \rightarrow U$ are functions.

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Suppose (S, d_S) , (T, d_T) , and (U, d_U) are metric spaces and $f : S \rightarrow T$ and $g : T \rightarrow U$ are functions.

If f is continuous at p and g is continuous at $f(p)$, then the composition $h = g \circ f : S \rightarrow U$ defined by $h(x) = g(f(x))$ is continuous at p .

Challenge!

Problem

Prove the previous theorem!

Preimages

Definition

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Definition

Suppose $f : S \rightarrow T$ is a function and that $Y \subseteq T$.
The **inverse image** or **preimage** of Y is the set

$$f^{-1}(Y) = \{x : x \in S, \text{ and } f(x) \in Y\}.$$

Challenge!

Problem

Suppose that $S \rightarrow T$ and $Y \subseteq Z \subseteq T$.
Prove that

$$f^{-1}(Y) \subseteq f^{-1}(Z).$$

Open sets

Theorem

Suppose (S, d_S) and (T, d_T) are metric spaces and $f : S \rightarrow T$ is a function.

Open sets

Theorem

Suppose (S, d_S) and (T, d_T) are metric spaces and $f : S \rightarrow T$ is a function.

Then f is continuous if and only if $f^{-1}(V)$ is open for all open subsets $V \subseteq T$.

Closed sets

Theorem

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Closed sets

Theorem

Suppose (S, d_S) and (T, d_T) are metric spaces and $f : S \rightarrow T$ is a function.

Then f is continuous if and only if $f^{-1}(A)$ is closed for all closed subsets $A \subseteq T$.